Final Strategy & Recommendation Report

Supply Chain Optimization Project

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This project analyzed e-commerce supply chain data to identify inefficiencies in delivery performance, demand planning, and inventory management. Using data-driven modeling techniques including time-series forecasting, EOQ optimization, safety stock calculation, and ABC classification, an integrated improvement strategy was developed.

The results show that forecasting-driven replenishment and EOQ-based ordering can significantly reduce inventory costs while improving service levels. Delivery delay risk was found to be regionally concentrated, enabling targeted logistics interventions. A dashboard was built to support continuous monitoring and decision-making.

The recommended strategy combines demand planning, differentiated inventory control, and logistics optimization to improve both operational efficiency and financial performance.

1. Key Findings from Analysis

Based on exploratory and modeling phases, the following key findings were established:

- Demand shows clear monthly and seasonal variation
- Category demand is highly skewed toward a small set of products
- Delivery delays are concentrated in specific regions and routes
- Forecast models can produce stable short-term demand projections
- EOQ-based ordering reduces combined holding and ordering costs
- ABC classification reveals that a small product set drives most revenue
- Safety stock is essential for high-variability products

These findings form the foundation for the final supply chain strategy.

2. Demand Planning Strategy

A structured demand planning framework is recommended.

Forecasting Policy

- Use rolling 30-day forecasts
- Update forecasts weekly for A-class products
- Update forecasts bi-weekly for B products
- Monthly forecast refresh for C products
- Use exponential smoothing for short-term operational planning

Forecast Governance

- Maintain forecast accuracy tracking (MAPE, MAE)
- Adjust forecasts for promotions and seasonal events
- Share forecasts with suppliers for capacity planning
- Maintain forecast vs actual dashboard

Expected benefit:

- Reduced demand uncertainty
- Better replenishment timing
- Lower stockout risk

3. Inventory Strategy

Inventory control should be differentiated using ABC classification.

3.1 A-Class Products (High Value)

Policy:

- Highest service level (98–99%)
- EOQ ordering
- Safety stock maintained
- Continuous review system
- Automated reorder triggers
- Weekly monitoring

Impact:

- Prevent revenue loss from stockouts
- Protect customer experience

3.2 B-Class Products (Medium Value)

Policy:

- EOQ ordering
- Moderate safety stock
- Periodic review (weekly/bi-weekly)

Impact:

- Balanced cost and service

3.3 C-Class Products (Low Value)

Policy:

- Bulk periodic ordering
- Low safety stock
- Monthly review
- Simplified control

Impact:

- Lower administrative effort
- Reduced ordering overhead

4. EOQ Implementation Strategy

EOQ-based replenishment should replace fixed-quantity ordering.

Implementation steps:

- Use annual demand estimates from forecast
- Use ordering cost = 100
- Use holding cost rate = 20%
- Compute EOQ per product
- Integrate EOQ values into ERP reorder settings

Expected outcomes:

- Lower total inventory cost
- Optimized order frequency
- Reduced excess stock

Estimated savings:

- Inventory cost reduction: **20 %**
- Annual savings: 12330.7

5. Safety Stock & Reorder Policy

Safety stock must be used for demand variability protection.

Policy:

- Use service-level-based safety stock
- Higher buffer for A products
- Lower buffer for C products
- Recalculate quarterly

Reorder point policy:

ROP = Lead Time Demand + Safety Stock

Impact:

- Reduced stockouts
- Improved service level
- Stable fulfillment rate

6. Implementation Roadmap

Phase-based rollout is recommended.

Phase 1 — ABC classification + KPI dashboard
Phase 2 — EOQ & safety stock deployment
Phase 3 — Forecast-driven replenishment
Phase 4 — Delivery optimization
Phase 5 — Regional warehouse strategy

Timeline: 3–6 months staged rollout.

7. Risk Mitigation Plan

Key risks and controls:

Demand spikes → safety stock buffers
Forecast error → weekly refresh
Lead time variability → dynamic ROP
Supplier delay → alternate vendors
Seasonality → seasonal forecast adjustment

Continuous KPI monitoring reduces operational risk.

Conclusion

This project demonstrates that data-driven supply chain management significantly improves operational efficiency and financial performance. Forecast-based demand planning, EOQ-driven replenishment, ABC-based control, and delivery risk targeting together form a robust optimization strategy.

The integrated dashboard enables continuous monitoring and supports data-backed decision making. Implementing the proposed strategy will improve service levels, reduce inventory costs, and strengthen supply chain resilience.